

selectivity and high-sensitivity sensors for biological and chemical detection. This advanced detection of harmful chemical and biological agents; micro sensors for radioactivity, low-power or self-powering consumption security electronics, polymer electronics, nano-optics will provide capabilities that are not available today. Another critical area is the development of nanofabrics. Nanofabrics have unique properties like decreased receptivity to chemical or biological agents, materials with the ability to expand and contract (like a thermostat) as to exhaust or conserve body heat, or to resist the penetration of a bullet.

Imagine, in a few years we will have advanced sensor networks that are self-powered and have the smarts to communicate from sensor to sensor and the ability to detect very small amounts of chemicals or biological agents installed in the water supplies across the country. Once a single sensor node detects the presence of one of the agents, it communicates to the others what hit was found and receives their verification. Once verified, the information is communicated to the control sensor that relays the information back for immediate action. Or imagine using it in underwater sensor networks to detect the movement of ships into and out of our ports and could detect chemical, biological or radiological materials in cargo containers.

One of the early Nanotechnology successes was material that had the characteristics of Gortex (used in bulletproof vests) with the look and feel of regular wool. Super strength nanofabrics sandwiched into normal contraction materials are expected to improve blast resistant construction practices making commercial and governmental structures much more resilient against bomb blasts. Bomb resistant containers for cargo and luggage on ships and aircraft, bomb resistant glass for office buildings and government complexes, advanced structural members that have the strength by are pliable to absorb the energy of a blast are all currently being investigated as real world application of nanotechnology.

Although nanotechnology is based in the research labs today, the advances made to date have illustrated the significant value this technology can bring. The enhanced characteristics of materials will allow us to create new and innovative devices to protect all of us from terrorism, both directly through safer construction designs and indirectly through intelligence and surveillance. With each passing day, the promise of nanotechnology, as our main weapon against terrorism, becomes more and more apparent.

Evolution into Cyborgs

According to Arthur C. Clarke Lifetime Achievement Award winner, futurist and inventor guy Ray Kurzweil, within the next 20-25 years, our mastery of nan-